Chapter 4

Expectations of Random Variables

4.1 Expected value

In this section, we introduce the concept of the *expected value* (or *expectation*) of a random variable X, denoted $\mathbb{E}(X)$. Informally, this represents the 'average value' of the random variable X. You may have seen this defined for discrete random variables before.

Definition 4.1. For a discrete random variable X, the *expected value* of X is

$$\boxed{\mathbb{E}(X) = \sum_{x \in \mathrm{Im}X} x \underbrace{p_X(x)}_{\text{pmf of } X} = \sum_{x \in \mathrm{Im}X} x\mathbb{P}(X = x)}$$

For a continuous random variable X, the *expected value* of X is

$$\boxed{\mathbb{E}(X) = \int_{-\infty}^{\infty} x \underbrace{f_X(x)}_{\text{pdf of } X} \mathrm{d}x = \int_{\mathrm{Im}X} x f_X(x) \mathrm{d}x}$$

Before we explore the mathematical definition let us consider different interpretations of what the expected value should represent.

Interpretations

Long run interpretation:
Suppose we run a very large number of simulations of X. We record the outcomes as $x_1, x_2, x_3, \ldots, x_n$. The proportion of times we see the value x we call $\hat{p}(x)$ and this should be approximately $p_X(x)$ given a large enough set of data. Therefore the average outcome, $\overline{x}$, is

$$\overline{x} = \sum_{i=1}^{n} x_i = \sum_{x \in \mathrm{Im}(X)} x\hat{p}(x) \approx \sum_{x \in \mathrm{Im}(X)} x p_X(x).$$

This will be justified using the law of large numbers in Chapter 8 at the end of the course.

Physical interpretation:
Suppose we have a rod of negligible weight with masses of size w_i placed at points x_i, for $i = 1, ..., n$. Then the centre of gravity of the rod is located at

$$\frac{\sum_{i=1}^{n} x_i w_i}{\sum_{i=1}^{n} w_i} = \sum_{i=1}^{n} x_i p_i.$$

Here $p_i = w_i / \sum w_i$ is the proportion of the mass located at x_i. Now if X is a random variable which takes the value x_i with probability p_i, then we can interpret the expected value of X as the centre of gravity of the rod.

Fair Price interpretation:
Suppose you pay a fixed amount £b to place a bet and receive a return of £X which is random. If you place n independent bets then your return is

$$S = X_1 + \ldots + X_n \approx \sum_{x \in \mathrm{Im}(X)} nx p_X(x) = n\mathbb{E}(X). \tag{4.1}$$

Then your average net return per bet is £$(\mathbb{E}(X) - b)$. So $\mathbb{E}(X)$ is the price of fair bet for a return of £X. This price makes wins and loses cancel out over the long run.

Remark 4.1.
- Compare the formula of expectation with the sample mean of frequency data, $\bar{x} = \sum_i \left(\frac{f_i}{n}\right) x_i$, where f_i is the frequency of observation x_i. The relative frequency $\frac{f_i}{n}$ for observation x_i is replaced by the probability $p_X(x)$ for $x \in \mathrm{Im}X$.
- $\mathbb{E}(X)$ is sometimes referred to as the *population mean* or *theoretical mean* in order to distinguish it from the *sample mean*.
- Observe the similarities in the definition of expectation for discrete and continuous random variables, with the main differences being the sum or integral and using the probability mass function or probability density function. Many of the comments we make about expectations will apply to both discrete and continuous random variables.

Example 2.5 continued. $X = \#$ heads in 3 flips of a fair coin. $\mathrm{Im}X = \{0, 1, 2, 3\}$. Probability mass function is

$$p_X(x) = \begin{cases} 1/8 & x = 0 \\ 3/8 & x = 1 \\ 3/8 & x = 2 \\ 1/8 & x = 3 \end{cases}$$

Expected value is

$$\mathbb{E}(X) = \sum_{x \in \mathrm{Im}X} x\, p_X(x) = 0 \times 1/8 + 1 \times 3/8 + 2 \times 3/8 + 3 \times 1/8 = 3/2$$

That is, when a fair coin is flipped 3 times, the expected number of Heads is 3/2 (unsurprisingly).

Example 4.1. Now suppose you play the following game: You pay £1 to select a number ω uniformly at random from $[0, 1]$. If the outcome $\omega > 0.5$, then you win £2; if the outcome $\omega \leq 0.5$, then you win nothing. Let $Z(\omega)$ be your net gain in this game.
The random variable Z can only take two values, i.e. $\mathrm{Im}(Z) = \{-1, 1\}$, so Z is a discrete random variable.
Probability mass function is

$$p_Z(z) = \begin{cases} 1/2 & z = -1 \\ 1/2 & z = 1 \end{cases}$$

Expected value is

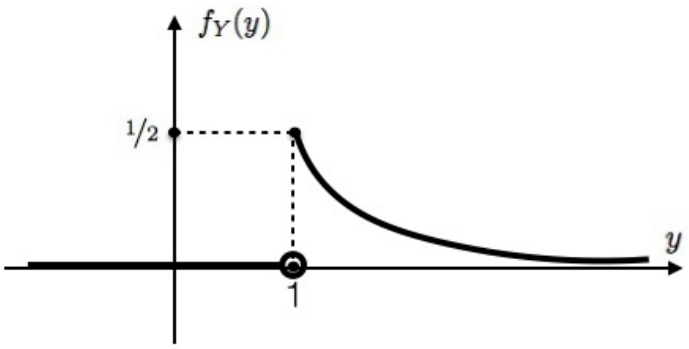
$$\mathbb{E}(X) = \sum_{x \in \mathrm{Im}X} x\, p_X(x) = -1 \times 1/2 + 1 \times 1/2 = 0$$

The expected net gain is zero, so we can say that the game is *fair*.

Example 4.2. Suppose Y is a random variable with density function given by

$$f_Y(y) = \begin{cases} \frac{2}{y^3} & \text{for } y \geqslant 1 \\ 0 & \text{otherwise} \end{cases}$$

The image of Y is $\mathrm{Im}Y = [1, \infty)$.



Then

$$\begin{aligned}\mathbb{E}(Y) &= \int_1^\infty y\, f_Y(y)\mathrm{d}y \\ &= \int_1^\infty y\frac{2}{y^3}\,\mathrm{d}y = \int_1^\infty \frac{2}{y^2}\,\mathrm{d}y = \left[\frac{-2}{y}\right]_1^\infty = \left(0 - \left(\frac{-2}{1}\right)\right) = 2.\end{aligned}$$

Warning: The expectation need not always exist as is shown in the following example.

Example 4.3 (Cauchy Distribution)**.** Let X be a continuous random variable with probability density function

$$f_X(x) = \frac{1}{\pi(1+x^2)}.$$

One can show that this is a valid probability density function however if the expectation exists it would be given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)}\,\mathrm{d}x.$$

Note that $\frac{x}{\pi(1+x^2)} \approx \frac{1}{x\pi}$ for x large and positive, and hence this function is not integrable.

Expected value of $g(X)$

Suppose X is a discrete random variable with probability mass function p_X, and that $Y = g(X)$ for some function $g : \mathbb{R} \to \mathbb{R}$. Then Y is itself a random variable, with its own probability mass function p_Y. If we want to compute the expected value of Y, then one way is to first find the probability mass function $p_Y(y)$, then use the definition

$$\mathbb{E}(Y) \;=\; \sum_{y\in \mathrm{Im}Y} y\, p_Y(y)$$

But there is a more direct way, without needing to explicitly find p_Y. If X is a discrete random variable with probability mass function p_X, and $Y = g(X)$ for some function $g : \mathbb{R} \to \mathbb{R}$, then

$$\boxed{\mathbb{E}(Y) = \mathbb{E}(g(X)) = \sum_{x\in \mathrm{Im}X} g(x)\, p_X(x).}$$

Similarly, if X is a continuous random variable with probability density function f_X, and $Y = g(X)$ for some function $g : \mathbb{R} \to \mathbb{R}$, then

$$\boxed{\mathbb{E}(Y) = \mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)\, f_X(x)\,\mathrm{d}x.}$$

Example 4.4. Fix some event A and let

$$g(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

Then

$$\mathbb{E}(g(X)) = \mathbb{P}(X \in A).$$

4.2 Properties of expected value

Proposition 4.1. *Let X be a random variable with finite expectation, and let $a, b \in \mathbb{R}$ be constant. Then the following properties hold:*

1. *The expected value of a constant: $\mathbb{E}(a) = a$.*
2. *Expectation is linear: $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$.*
3. *If X is non-negative (i.e. $\mathbb{P}(X \geq 0) = 1$), then $\mathbb{E}(X) \geq 0$.*

Proof We will establish these properties for continuous random variables, the proof for discrete random variables is analogous and in fact these properties hold for any random variables with finite expectation.
Let X be a continuous random variable with probability density function f_X be such that $\mathbb{E}[X] < \infty$, and let $a, b \in \mathbb{R}$ be constants.

1.

$$\mathbb{E}(a) = a \times \int_{-\infty}^{\infty} f_X(x)\mathrm{d}x = a \times 1 = a.$$

2.

$$\begin{aligned} \mathbb{E}(aX + b) &= \int_{-\infty}^{\infty} (ax + b) f_X(x)\mathrm{d}x \\ &= a\int_{-\infty}^{\infty} x f_X(x)\mathrm{d}x + b\int_{-\infty}^{\infty} f_X(x)\mathrm{d}x \\ &= a\mathbb{E}(X) + b \times 1 = a\mathbb{E}(X) + b \end{aligned}$$

3. Since X is non-negative, we have $f_X(x) = 0$ for all $x < 0$ therefore

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\mathrm{d}x = \int_{0}^{\infty} x f_X(x)\mathrm{d}x.$$

From the definition of a probability density function we have $f_X(x) \geq 0$ for all $x \in \mathbb{R}$. Hence $x f_X(x) \geq 0$ for all $x \geq 0$ and using that the integral of a non-negative function is non-negative we have

$$\mathbb{E}[X] = \int_{0}^{\infty} x f_X(x)\mathrm{d}x \geq 0.$$

□

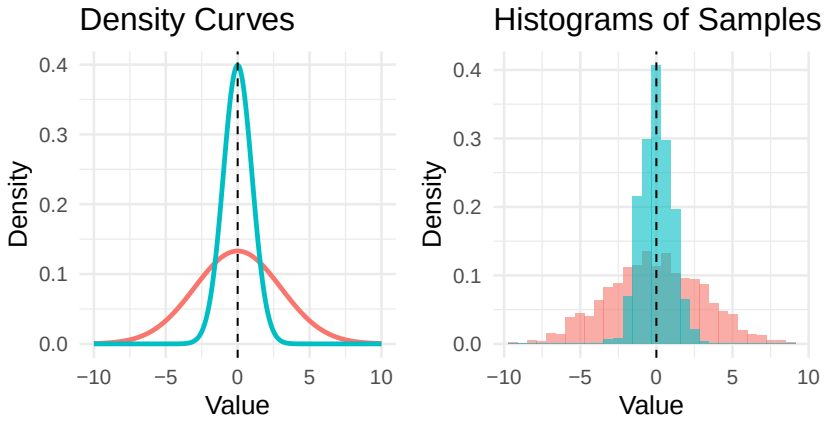


Figure 4.1: The left panel shows the theoretical density curves for two normal distributions with the same mean (indicated by the dashed black line) but different variances: the blue curve corresponds to the distribution with low variance ($\sigma^2 = 1$), and the orange curve to the distribution with high variance ($\sigma^2 = 9$). The right panel displays histograms of samples drawn from these distributions, illustrating how increased variance leads to greater spread in observed values.

4.3 Variance of a random variable

Once we understand expectation as a measure of the "average" or "central" value of a random variable, it's natural to ask how much the values of that variable tend to deviate from this average. This is where variance comes in. Variance quantifies the spread or dispersion of a random variable's values around its expectation. In other words, it tells us how "spread out" the outcomes are, giving insight into the variability or uncertainty inherent in the random process.

Definition 4.2. The variance of a random variable X is the expected value of its *squared deviation* from its expected value $\mathbb{E}(X)$:

$$\boxed{\mathbb{V}\text{ar}(X) = \mathbb{E}\left((X - \mathbb{E}(X))^2\right) = \mathbb{E}\left((X-\mu)^2\right) \quad \text{where } \mu = \mathbb{E}(X)}$$

Notation: We sometimes write σ_X^2 for $\mathbb{V}\text{ar}(X)$

Remark 4.2. To understand the form of the variance formula, recall that variance is meant to measure how far values of a random variable deviate from the mean, i.e. the spread of $X - \mathbb{E}[X]$. Simply averaging the deviations from the mean would always give zero: Since positive and negative deviations cancel out, that is

$$\mathbb{E}\left[X - \mathbb{E}[X]\right] = 0.$$

To avoid this, we square the deviations, ensuring all contributions are non-negative and that larger deviations are penalized more heavily. The variance is then defined as the expected value of these squared deviations. This captures the average squared distance from the mean, providing a natural measure of spread in the distribution.

While variance gives us a measure of spread, it does so in squared units. For example, if X is measured in metres, then $\mathbb{V}\text{ar}(X)$ is in square metres.

This can make interpretation less intuitive. To bring the measure of spread back to the original units of the data, we take the square root of the variance.

This gives us the standard deviation:

$$\boxed{\sigma_X = \sqrt{\mathbb{V}\text{ar}(X)}}$$

The standard deviation is often more interpretable in practice, as it tells us, on average, how far values tend to deviate from the mean — in the same units as the data itself.

Properties of $\mathbb{V}\text{ar}(X)$

Proposition 4.2. *Let X be a random variable with finite second moment, i.e. $\mathbb{E}[X^2] < \infty$, and let $a, b \in \mathbb{R}$ be constant. Then the following properties hold:*

1. *The variance is non-negative:* $\mathbb{V}\text{ar}(X) \geq 0$
2. *The variance of constants is zero:* $\mathbb{V}\text{ar}(a) = 0$
3. *Variance is shift independent but scales quadratically:*

$$\mathbb{V}\text{ar}(aX + b) = a^2\mathbb{V}\text{ar}(X)$$

Special cases of Property 3*:*

$\mathbb{V}\text{ar}(aX) = a^2\mathbb{V}\text{ar}(X)$
$\mathbb{V}\text{ar}(X + b) = \mathbb{V}\text{ar}(X)$

4. *A useful formula for computations:*

$$\mathbb{V}\text{ar}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Proof

1. Since $(X - \mu)^2 \geq 0$, by Property 3 of expected values, we have

$$\mathbb{V}\text{ar}(X) = \mathbb{E}((X - \mu)^2) \geq 0$$

2. Since $\mathbb{E}(a) = a$ all the deviations are zero, i.e. $a - \mathbb{E}[a] = 0$.
Hence the squared deviations are zero and using that $\mathbb{E}[0] = 0$ we have $\mathbb{V}\text{ar}(a) = \mathbb{E}((a - \mathbb{E}(a))^2)) = \mathbb{E}(0) = 0$.

3.

$$\begin{aligned}
\mathbb{V}\text{ar}(aX + b) &= \mathbb{E}((aX + b - \mathbb{E}(aX + b))^2) \\
&= \mathbb{E}((aX + b - a\mathbb{E}(X) - b)^2) \qquad \text{(by the linearity of expected value)} \\
&= \mathbb{E}(a^2(X - \mathbb{E}(X))^2) \qquad \text{(by algebra)} \\
&= a^2\mathbb{E}((X - \mathbb{E}(X))^2) \qquad \text{(by the linearity of expected value)} \\
&= a^2\mathbb{V}\text{ar}(X) \qquad \text{(by the definition of } \mathbb{V}\text{ar}(X)
\end{aligned}$$

4. Let $\mu = \mathbb{E}[X]$ then

$$\begin{aligned}
\mathbb{V}\text{ar}(X) = \mathbb{E}((X - \mu)^2) &= \mathbb{E}(X^2 - 2\mu X + \mu^2) \\
&= \mathbb{E}(X^2) - 2\mu\mathbb{E}(X) + \mu^2 \\
\mathbb{V}\text{ar}(X) &= \mathbb{E}(X^2) - \mu^2 \quad (\text{since } \mathbb{E}(X) = \mu)
\end{aligned}$$

□

In practise it is often easier to compute the variance using Property 4 in Proposition 4.2 rather than calculating the variance directly.

Example 2.5 continued. $X = \#$ heads in 3 flips of a fair coin. $\text{Im}X = \{0, 1, 2, 3\}$. Probability mass function is

$$p_X(x) = \begin{cases} 1/8 & x = 0 \\ 3/8 & x = 1 \\ 3/8 & x = 2 \\ 1/8 & x = 3 \end{cases}$$

Find the variance for X, $\mathbb{V}\text{ar}(X)$.

Solution:
From the earlier calculations, we found $\mathbb{E}(X) = \frac{3}{2}$.

$$\mathbb{E}(X^2) = \sum_{x \in \text{Im}(X)} x^2 p_X(x) = 0^2(1/8) + 1^2(3/8) + 2^2(3/8) + 3^2(1/8) = 3.$$

Therefore

$$\begin{aligned} \mathbb{V}\text{ar}(X) &= \mathbb{E}(X^2) - (\mathbb{E}(X))^2 \\ &= 3 - \left(\frac{3}{2}\right)^2 \\ &= \frac{3}{4}. \end{aligned}$$

4.4 Expectation and Variance of important examples

4.4.1 Bernoulli Random Variable

Recall $X \sim \text{Ber}(p)$ if and only if X is a discrete random variable with $\text{Im}(X) = \{0, 1\}$ and $p = \mathbb{P}(X = 1)$. Then it follows that

$$\mathbb{E}(X) = 0 \times \mathbb{P}(X = 0) + 1 \times \mathbb{P}(X = 1) = p$$

Also,

$$\mathbb{E}(X^2) = 0^2 \times \mathbb{P}(X = 0) + 1^2 \times \mathbb{P}(X = 1) = p,$$

and so

$$\mathbb{V}\text{ar}(X) = \mathbb{E}(X^2) - \big(\mathbb{E}(X)\big)^2 = p - p^2 = p(1 - p)$$

4.4.2 Binomial Random Variable

Recall $X \sim \text{Bin}(n, p)$ if and only if X is a discrete random variable with $\text{Im}(X) = \{0, 1, \ldots, n\}$ and

$$p_X(k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k \in \{0, 1, \ldots, n\}.$$

Given the probability mass function $p_X(x)$, we can find the expected value of $X \sim \text{Bin}(n,p)$:

$$\begin{aligned}
\mathbb{E}(X) = \sum_{k=0}^{n} k p_X(k) &= \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} \\
&= \sum_{k=0}^{n} k \left(\frac{n!}{k!\,(n-k)!} \right) p^k (1-p)^{n-k} && \text{by definition of } \binom{n}{k} \\
&= \sum_{k=1}^{n} \frac{n!}{(k-1)!\,(n-k)!}\, p^k (1-p)^{n-k} && \text{since } k! = k \times (k-1)! \\
&= np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!\,(n-k)!}\, p^{k-1} (1-p)^{n-k} && \text{since } n! = n \times (n-1)! \\
&= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{(n-1)-j} && \text{changing the variable } j = k-1
\end{aligned}$$

Notice that the summation in the final line above is the sum of all terms in the probability mass function of the $\text{Bin}(n-1,p)$ distribution, so must sum to 1, and so we obtain

$$\mathbb{E}(X) = np$$

That is, if we carry out n trials, each with success probability p, the expected number of successes is np.

We will show later that for $X \sim \text{Bin}(n,p)$,

$$\mathbb{V}\text{ar}(X) = np(1-p).$$

4.4.3 Geometric Random Variable

Recall $X \sim \text{Geo}\,(p)$ if and only if X is a discrete random variable with $\text{Im}(X) = \{1,2,3,\ldots\}$ and

$p_X(k) = p(1-p)^{k-1}, k \in \{1,2,3,\ldots\}$.

Now find the expectation of $X \sim \text{Geo}\,(p)$:

$$\mathbb{E}(X) \quad = \quad \sum_{x \in \text{Im}X} x p_Y(x) = \sum_{x=1}^{\infty} x(1-p)^{x-1} p = p \sum_{x=1}^{\infty} x(1-p)^{x-1}$$

To evaluate this infinite sum we use a geometric series. Recall that the sum of a geometric series is given by

$$\sum_{x=0}^{\infty} a^x = \frac{1}{1-a}, \qquad \text{for any } a \text{ such that } |a| < 1.$$

This is not quite of the form we need, however if we differentiate both sides with respect to a we have

$$\sum_{x=1}^{\infty} x a^{x-1} = \frac{1}{(1-a)^2}, \qquad \text{for any } a \text{ such that } |a| < 1.$$

Note here the term corresponding to $x = 0$ was constant therefore once we differentiate it vanishes. Then taking $a = 1-p$, then provided $p < 1$ we obtain

$$\mathbb{E}(X) \quad = \quad p \left(\frac{1}{(1-(1-p))^2} \right) = \frac{p}{p^2} = \frac{1}{p}$$

That is, for $X \sim \text{Geo}(p)$ with $0 \leqslant p < 1$,

$$\mathbb{E}(X) = \frac{1}{p}$$

(Exercise: find $\mathbb{E}(X)$ when $p = 1$. Hint: before doing any calculations, think about the experiment in this case.)

Exercise: Show that for $X \sim \text{Geo}(p)$,

$$\mathbb{V}\text{ar}(X) = \frac{1-p}{p^2}$$

Hint: Note that for $-1 < x < 1$,

$$\frac{1}{(1-x)^2} = \frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{d}{dx}\sum_{k=0} x^k = \sum_{k=1}^{\infty} kx^{k-1},$$
$$\frac{2}{(1-x)^3} = \frac{d^2}{dx^2}\left(\frac{1}{1-x}\right) = \frac{d^2}{dx^2}\sum_{k=0}^{\infty} = \sum_{k=2}^{\infty} k(k-1)x^{k-2}.$$

4.4.4 Poisson Random Variable

Recall that $X \sim \text{Poi}(\lambda)$ if and only if X is a discrete random variable, with $\text{Im}(X) = \{0, 1, 2, 3, ...\}$ and

$$p_X(x) = \frac{\mathrm{e}^{-\lambda}\lambda^x}{x!}, x \in \{0, 1, 2, 3, \ldots\}.$$

To find the expected value:

$$\begin{aligned}
\mathbb{E}(X) = \sum_{x \in \text{Im}(X)} x p_X(x) &= \sum_{x=0}^{\infty} x\left(\frac{\lambda^x}{x!}\,\mathrm{e}^{-\lambda}\right) \\
&= \mathrm{e}^{-\lambda}\sum_{x=1}^{\infty}\frac{\lambda^x}{(x-1)!} \\
&= \lambda\mathrm{e}^{-\lambda}\sum_{y=0}^{\infty}\frac{\lambda^y}{y!} \text{ where } y = x-1 \\
&= \lambda\mathrm{e}^{-\lambda}\mathrm{e}^{\lambda} \\
\mathbb{E}(X) &= \lambda
\end{aligned}$$

Recall that in deriving the Poisson probability mass function, we originally set λ to be 'the expected number of events'. We have just checked, from the probability mass function and the formula for expected value, that indeed $\mathbb{E}(X) = \lambda$.

It can be shown similarly that the variance of $X \sim \text{Poi}(\lambda)$ is

$$\mathbb{V}\text{ar}(X) = \lambda$$

(Exercise: Show that $\mathbb{V}\text{ar}(X) = \lambda$. Hint: start by finding $\mathbb{E}(X(X-1))$ for $X \sim \text{Poi}(\lambda)$.)

The Poisson distribution has the distinctive property that the mean and variance are the same.

4.4.5 Uniform Random Variable

Recall that $X \sim \mathrm{U}(a,b)$ if and only if X is a continuous random variable with

$$f_X(x) = \frac{1}{b-a}\mathbb{I}_{[a,b]}(x).$$

Find the expected value:

$$\begin{aligned}\mathbb{E}(X) &= \int_{-\infty}^{\infty} x f_X(x)\,dx \\ &= \int_a^b x\left(\frac{1}{b-a}\right)dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b = \frac{1}{b-a}\left(\frac{b^2-a^2}{2}\right) = \frac{(b-a)(b+a)}{2(b-a)} = \frac{a+b}{2}\end{aligned}$$

Notice that this expected value is unsurprising.

Can also show (*exercise*) that $\mathbb{V}\mathrm{ar}(X) = \frac{(b-a)^2}{12}$

4.4.6 Exponential Random Variables

Recall $X \sim \mathrm{Exp}(\lambda)$ if and only if X is a continuous random variable with pdf

$$f_X(x) = \begin{cases}\lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0.\end{cases}$$

$$\begin{aligned}\mathbb{E}(X) = \int_0^\infty x f_X(x)\mathrm{d}x = \int_0^\infty \lambda x e^{-\lambda x}\mathrm{d}x \\ &= \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x}\mathrm{d}x \qquad \text{(integrate by parts)} \\ &= \left[-xe^{-\lambda x}\right]_0^\infty - \left[\frac{e^{-\lambda x}}{\lambda}\right]_0^\infty \\ &= \left(-(0)e^{-\lambda(0)} - 0\right) - \left(\frac{e^{-\lambda(0)}}{\lambda} - 0\right) \\ &= \frac{1}{\lambda}\end{aligned}$$

(Note: You can use the exponential series to show $\lim_{M\to\infty} Me^{-\lambda M} = \lim_{M\to\infty} \frac{M}{e^{\lambda M}} = 0$, indeed,

$$0 \leq \frac{M}{e^{\lambda M}} = \frac{M}{1+\lambda M + \frac{1}{2}\lambda^2 M^2 + \ldots} \leq \frac{M}{1+\lambda M + \frac{1}{2}\lambda^2 M^2} \to 0$$

as $M \to \infty$.)

Exercise: Show that $\mathbb{V}\mathrm{ar}(X) = \frac{1}{\lambda^2}$

4.4.7 Gamma Random Variable

Recall $X \sim \mathrm{Gamma}\,(\alpha,\lambda)$ if and only if X is a continuous random variable with probability density function

$$f_X(x) = \begin{cases}\frac{\lambda^\alpha}{\Gamma(\alpha)} e^{-\lambda x} x^{\alpha-1}, & x > 0, \\ 0, & x \leq 0.\end{cases} \tag{4.2}$$

We can use this to find $\mathbb{E}(X)$ and $\mathbb{V}\text{ar}(X)$, as follows.

$$\begin{aligned}\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x)\,dx &= \int_0^{\infty} x\left(\frac{\lambda^\alpha}{\Gamma(\alpha)}\,x^{\alpha-1}e^{-\lambda x}\right)dx \\ &= \frac{\lambda^\alpha}{\Gamma(\alpha)}\int_0^{\infty} x^\alpha e^{-\lambda x}\,dx\end{aligned}$$

Making the substitution $y = \lambda x$ in the integral,

$$\begin{aligned}\mathbb{E}(X) &= \frac{\lambda^\alpha}{\Gamma(\alpha)}\int_0^{\infty}\left(\frac{y}{\lambda}\right)^\alpha e^{-y}\,\frac{dx}{\lambda} \\ &= \frac{1}{\lambda\Gamma(\alpha)}\int_0^{\infty} y^\alpha e^{-y}\,dy \\ &= \frac{1}{\lambda\Gamma(\alpha)}\Gamma(\alpha+1) \qquad \text{(from definition of Gamma function)} \\ &= \frac{\alpha}{\lambda} \qquad \text{(since } \Gamma(\alpha+1) = \alpha\Gamma(\alpha)\text{)}\end{aligned}$$

Exercise: Show that $\mathbb{V}\text{ar}(X) = \frac{\alpha}{\lambda^2}$